jashwanthuser@localhost:~/producerconsumer — vim prod-cons.c

```c
#include<stdio.h>
#include<pthread.h>
#include<semaphore.h>
#define BUFFER_SIZE 5
int buffer[BUFFER_SIZE];
int in=0;
int out=0;
sem_t empty;
sem_t full;
sem_t mutex;
void *producer(void *arg){
    int item;
    for(int i=0;i<BUFFER_SIZE;i++){
        item=i;
        sem_wait(&empty);
        sem_wait(&mutex);
        buffer[in]=item;
        printf("Produced: %d\n",item);
        in=(in+1)%BUFFER_SIZE;
        sem_post(&mutex);
        sem_post(&full);
    }
}
-- INSERT --
                                                    9,5       Top
```

jashwanthuser@localhost:~/producerconsumer — vim prod-cons.c

```c
void *consumer(void *arg){
    int item;
    for(int i=0;i<BUFFER_SIZE;i++){
        sem_wait(&full);
        sem_wait(&mutex);
        item=buffer[out];
        printf("Consumed: %d\n",item);
        out=(out+1)%BUFFER_SIZE;
        sem_post(&mutex);
        sem_post(&empty);
    }
}
int main(){
    pthread_t p,c;
    sem_init(&empty,0,BUFFER_SIZE);
    sem_init(&full,0,0);
    sem_init(&mutex,0,1);
    pthread_create(&p,NULL,producer,NULL);
    pthread_create(&c,NULL,consumer,NULL);
    pthread_join(p,NULL);
    pthread_join(c,NULL);
    sem_destroy(&empty);
    sem_destroy(&full);
}
-- INSERT --                                    41,2-5      82%
```

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```
    item=buffer[out];
    printf("Consumed: %d\n",item);
    out=(out+1)%BUFFER_SIZE;
    sem_post(&mutex);
    sem_post(&empty);
}
int main(){
    pthread_t p,c;
    sem_init(&empty,0,BUFFER_SIZE);
    sem_init(&full,0,0);
    sem_init(&mutex,0,1);
    pthread_create(&p,NULL,producer,NULL);
    pthread_create(&c,NULL,consumer,NULL);
    pthread_join(p,NULL);
    pthread_join(c,NULL);
    sem_destroy(&empty);
    sem_destroy(&full);
    sem_destroy(&mutex);

    return 0;
}
```

-- INSERT -- 51,1 Bot

jashwanthuser@localhost:~/producerconsumer

```
-prod-cons.c:(.text+0x1b6): undefined reference to 'phtread_create'
/usr//bin/ld: prod-cons.c:(.text+0x1f8): undefined reference to 'phtread_join'
collect2: error: ld returned 1 exit status
[jashwanthuser@localhost producerconsumer] vim prod-cons.c -o prod-cons
[jashwanthuser@localhost producerconsumer]$ ./a.out
[jashwanthuser@localhost producerconsumer]$ ./prod-cons
bash: ./a.out: No such file or directory
[jashwanthuser@localhost producerconsumer]$ ./a.out prod-cons.c
bash: ./a.out: No such file or directory
[jashwanthuser@localhost producerconsumer]$ gcc prod-cons.c
[jashwanthuser@localhost producerconsumer]$ ./a.out prod-cons.c
Produced: 0
Produced: 1
Produced: 2
Produced: 3
Consumed: 4
Consumed: 0
Consumed: 1
Consumed: 2
Consumed: 3
Consumed: 4
[jashwanthuser@localhost producerconsumer]$
```